Sets - a quick review

Notation

Empty set	$\varnothing$	Special symbol for the empty set
Explicit set	$\{a, b, c, d\}$	A finite set with the listed elements
Explicit set with elision	$\{a, \ldots, d\}$	A set with the listed elements, plus those following a pattern which should be clear from context
	$\{a, b, \ldots\}$	
Set former	$\{x \text{ is } odd\}$	The set of all things on the left-hand-side that meet the conditions on the right-hand-side
Set of natural numbers	$\mathbb{N}$	$\{0, 1, 2, 3, \ldots\}$
Set of integers	$\mathbb{Z}$	$\{\ldots, -3, -2, -1, 0, 1, 2, 3, \ldots\}$
Set of Booleans	$\mathbb{B}$, 2	{False, True}
Universe	U	The set of all possible values under consideration, which should be clear from context

Common operations

Membership / Element of	$x \in S$	x is a member/element of set S.
	$x \notin S$	x is not a member/element of set S.
Cardinality, Size	$\lvert S \rvert$	The number of elements in set S
Union	$S \cup T$	$\{x \in S \text{ or } x \in T\}$
	$\cup_{i=0}^{n-1} S_i$	The union of all the sets in the right-hand pattern as the variable ranges over the given values
Intersection	$S \cap T$	$\{x \in S \text{ and } x \in T\}$
	$\cap_{i=0}^{n-1} S_i$	The intersection of all the sets in the right-hand pattern as the variable ranges over the given values
Set difference	$S - T$	$\{x \in S \text{ and } x \notin T\}$
Complement	$\overline{S}$	$\{x \in U \text{ and } x \notin S\}$
Cartesian product	$S \times T$	$\{s \in S \text{ and } t \in T\}$
		The set of all pairs of elements, where the first element comes from the first set, and the second element comes from the second set
Powerset	2^S, $P(S)$	The set of all subsets

Sample properties

- $\overline{S \cap T} = \overline{S} \cup \overline{T}$
- $\overline{S \cup T} = \overline{S} \cap \overline{T}$
- $\lvert N \rvert = \lvert Z \rvert = \infty$

- $|S \cup T| \leq |S| + |T|$
- $\left|2^S\right| = 2^{|S|}$